Longchang Liu

Software Engineer

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Education

Hangzhou Dianzi University B.S. in Computer Science and Technology Aug 2022 - Jun 2026

Experience

Siorigin - Agentic Infra (Kernel-Gen Agent) Jul 2026 - Present

An agentic infrastructure system for internal workloads: an agent autonomously generates, validates and tunes GPU kernels, and the system exposes a unified kernel generation and evaluation capability internally — reaching the published state of the art on FlashInfer benchmarks.

- **Generation**: led the end-to-end loop — the agent reads a target operator, writes candidate kernels, and iterates until a version passes correctness and beats the baseline.
- **Evaluation**: each candidate is compiled, checked against a PyTorch reference and profiled on the target GPU, then scored on measured latency and achieved bandwidth; both gates must pass before delivery.
- **Search and feedback**: profiler output is attributed to memory- or compute-bound and fed back as the next prompt, keeping only useful evidence and failure reasons to bound context growth, with iteration caps and regression rejection holding the search monotonic.
- **Delivery**: internal users only supply the target operator and a reference implementation to get back a validated, benchmarked kernel, with no GPU programming or tuning on their side.

Alibaba - Amap (Intern) Oct 2025 - Jan 2026

Responsible for productionizing the Order Analysis Assistant and AI Ride-Hailing LLM projects, improving complex task handling and integrating shared AI capabilities.

- On the Order Analysis Assistant, designed and launched Deep Research, improving complex task decomposition and handling stability in long-context scenarios.
- Participated in the cold start of AI Ride-Hailing, building unified and stable AI invocation interfaces to keep the core path reliably integrated.

Tencent - Ads ChatBI System (Intern) Apr 2025 - Sep 2025

Contributed to the NL2Chart core workflow of Tencent Ads ChatBI, enabling conversion from natural-language requests to charts.

- To guarantee source corpus quality, designed and implemented an automated corpus review system, raising core-workflow recall accuracy from 50% to 80%; also introduced a state machine for stability and crash recovery.

Open Source

LangChain4j (12k+ stars) contributor — Java ecosystem AI framework; community issue fixes and capability improvements.

Technical Skills

- **AI Infra**: Transformer architecture and KV Cache; agent context management, multi-agent collaboration patterns, harness design and eval systems, with hands-on GPU kernel generation and tuning.
- **CS Fundamentals**: Solid OS and networking fundamentals — memory management (page tables), system calls, thread scheduling, TCP/UDP and congestion control.
- **Java & Engineering**: JVM class loading, memory model and G1/ZGC; Synchronized, Volatile, AQS and thread-pool internals; SQL optimization, MySQL tuning and production troubleshooting.

刘龙昌

软件开发工程师

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教育经历

杭州电子科技大学 计算机科学与技术 · 本科 2022 年 08 月 - 2026 年 06 月

工作经历

紫荆芯界 - Agentic Infra（Kernel-Gen Agent） 2026 年 07 月 - 至今

面向内部业务负载的 agentic infra 系统：由 agent 自主完成 GPU kernel 的生成、校验与调优，并向内部统一提供 kernel 生成与评测能力，在 FlashInfer benchmark 上达到业界 SOTA 水平。

- **生成链路**：主导从生成到调优的完整闭环——agent 读取目标算子、生成候选 kernel 并持续迭代，直到通过正确性校验且性能超过基线。
- **评测体系**：候选 kernel 经编译、与 PyTorch 参考实现数值对拍、GPU profiling 后，按实测延迟与有效带宽打分，两道关卡都过才允许交付。
- **搜索与反馈**：把 profiling 结果按 memory-bound / compute-bound 归因后回灌为下一轮 prompt，只保留有效证据与失败原因以控制上下文膨胀，并配合迭代上限与性能回退拒绝，让长时间搜索保持单调。
- **内部交付**：业务侧只需给出目标算子与参考实现，即可拿到经过正确性校验与性能评估的 kernel，无需自行处理 GPU 编程与调优细节。

阿里巴巴 - 高德地图（实习） 2025 年 10 月 - 2026 年 01 月

负责“订单分析助手”与“AI 打车”的大模型工程化落地，提升复杂任务处理能力并完成统一 AI 能力接入。

- 在订单分析助手项目中，设计并落地 Deep Research 功能，提升复杂任务拆解效果与长上下文场景下的处理表现。
- 参与 AI 打车服务冷启动，建设统一、稳定的 AI 调用接口，保障核心链路稳定接入。

腾讯 - 广告 ChatBI 系统（实习） 2025 年 04 月 - 2025 年 09 月

参与腾讯广告 ChatBI 的 NL2Chart 核心链路建设，支撑自然语言请求到图表生成的转化。

- 为了保证源头语料库的质量，设计并实现语料自动化评审系统，推动核心链路召回正确率从 50% 提升至 80%。同时引入状态机，增强服务稳定性，并提供宕机恢复能力。

开源经历

LangChain4j（12k+ Star）贡献者，Java 生态 AI 开发框架，参与社区问题修复与能力完善。

专业技能

- **AI Infra**：理解 Transformer 架构与 KV Cache；熟悉 Agent 上下文管理、Multi-Agent 协作范式、harness 设计与 eval 体系，有 GPU kernel 生成与调优实践
- **计算机基础**：扎实的操作系统与网络基础，理解内存管理（页表）、System Call、线程调度、TCP/UDP 与拥塞控制
- **Java 与工程实践**：熟悉 JVM 类加载机制、内存模型与 G1/ZGC，理解 Synchronized、Volatile、AQS 与线程池实现；有 SQL 优化、MySQL 调优与线上故障排查经验